

Radio latency and correction-burst findings

race-tracker · XBee-PRO 900HP (S3B) shared radio · measurements of 9 October 2026 · one Mac, two radios on USB-serial adapters, RTK base running

Summary

- **A telemetry transmission conflicts with the correction burst only in a short window.** Frames written from about 40 ms before the burst start to about 10-20 ms after it were delayed by 20-35 ms (or, rarely, lost), and in that window 46% of correction bursts arrived incomplete (19 of 41), against 1.8% (2 of 114) otherwise. Outside it, latency is normal.
- **The burst itself is short and regular:** every 1000 ms, median 28 ms wide, 95th percentile 31 ms, widest 33 ms (focus run).
- **Latency is steady when clear of the burst:** 13.0 ms median for a 26-byte frame, with a 90th percentile of 13.3 ms. About 1-2% of frames take 25-35 ms longer than that, and one in 600 took 306 ms.
- **Delay grows predictably with frame size:** about 7.6 ms plus 0.21 ms per byte (26 bytes: 13.0 ms; 84 bytes: 25.1 ms).
- **Consequence for the schedule:** block telemetry writes from 40 ms before to 20 ms after the burst start, plus a guard - roughly 80 ms per burst, not the 180-276 ms assumed earlier. That leaves about 92% of each 1 s cycle (96% of a 2 s cycle) for telemetry.
- **Not yet covered:** a Raspberry Pi's own delays, other adapter types, and the radio's on-air time per packet (estimated, not measured).

What was measured

`npm run radio-latency` opens two radios on the same machine. One sends frames at known times; the other decodes them. Both ends use the same clock, so the time from the app's `send()` to the other radio's decoded frame needs no clock synchronisation. The listener also timestamps each RTCM correction burst (the first RTCM message after a quiet gap), so every test frame can be placed relative to the burst. The delay includes the serial write, the radio's packetizing and RF time, the serial read and USB/Node scheduling - everything that makes a slot's start uncertain.

Run	Frame	Frames	Settings
A	26-byte position frame	300	default; sends spread over the whole second
B	84-byte batch frame (4 fixes)	600	LATENCY_BATCH=4 LATENCY_FRAMES=600
C (focus)	26-byte position frame	150	LATENCY_FOCUS=1: one frame per burst, swept from 80 ms before to 160 ms after the burst start

Results

	A: 26 B	B: 84 B	C: 26 B focus
Frames sent / received / lost	300 / 293 / 7 (2.3%)	600 / 589 / 11 (1.8%)	150 / 149 / 1 (0.7%)
Latency min / p50 (ms)	12.3 / 13.0	24.6 / 25.1	12.3 / 12.9
Latency p90 / p99 / max (ms)	13.3 / 36.1 / 45.4	25.3 / 49.1 / 306.5	31.4 / 48.2 / 52.5
Frames clear of the burst: p50 / p90 / p99 (ms)	13.0 / 13.2 / 15.9	25.1 / 25.3 / 49.1	12.9 / 13.3 / 48.2*
Lost, near the burst	7 of 23	7 of 185**	1 of 49
Lost, clear of the burst	0 of 276	4 of 412**	0 of 101

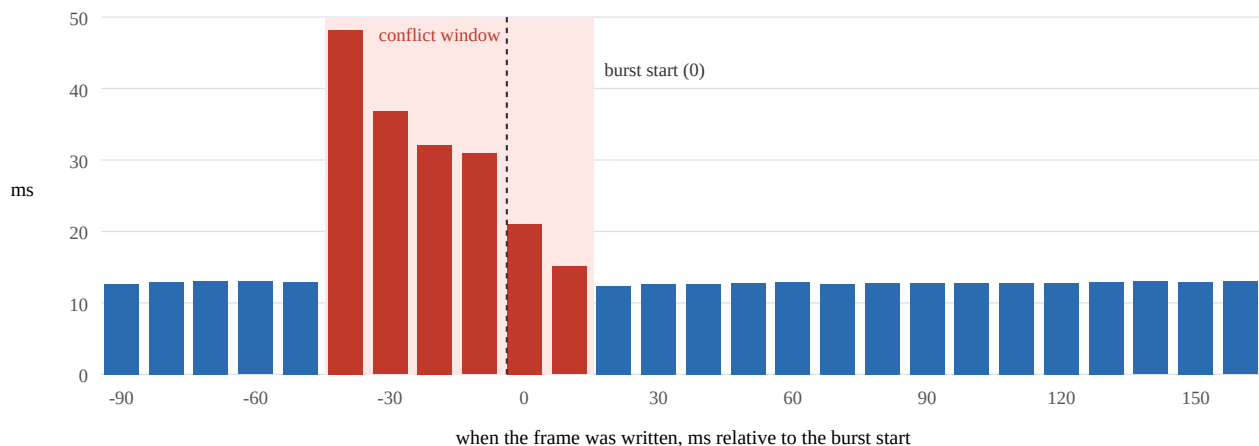
	A: 26 B	B: 84 B	C: 26 B focus
Bursts incomplete: with a test frame near / without	4 of 18 (22%) / 3 of 66 (4.5%)	9 of 33 (27%) / 6 of 131 (4.6%)	19 of 41 (46%) / 2 of 114 (1.8%)
Burst width (ms)	up to 33	median not reported; widest 273	median 28, p95 31, widest 33

* Includes two frames sent 30-40 ms before a burst, which the run's classification called "clear"; the phase table below shows they were delayed. ** Run B classified "near the burst" using its widest burst (273 ms), which made the window about 300 ms wide, so its near/clear split is diluted. The script now uses fixed edges measured in run C (-45 to +25 ms around the burst start).

Delay against frame size

Between the 26-byte and 84-byte runs the median rose 12.1 ms for 58 more bytes: a fixed part of about **7.6 ms** plus about **0.21 ms per byte**. Two passes through a 115200-baud serial line (write and read) account for 0.17 ms per byte; the rest is about the radio's air time at 200 kb/s (0.04 ms per byte). For a batch frame the data reaches the far end about 25 ms after the write, and the schedule can subtract that offset. Only the variation needs to go into the guard.

Median latency (ms) by when the frame was written - run C



10 ms buckets, 2-8 frames each. Latency is about 13 ms except in the red buckets, where the radio held the frame back (31-48 ms) and one frame was lost (0 to 10 ms).

Implications for the schedule

- **Blocked window:** hold writes from 40 ms before to 20 ms after the burst start (the first RTCM message the radio decodes), then add a guard of about 10 ms each side: roughly **80 ms per burst**.
- **Anchor on the burst start, as heard by the same radio,** and re-anchor every cycle. Two radios hearing the same bursts agreed within 0-3 ms (earlier two-listener capture); the base's own burst-to-burst timing wanders by about 5 ms.
- **Frames are delayed more than lost** near the burst (1 of 49 lost in run C), but the corrections are the casualty: transmitting in the window is what leaves bursts incomplete.
- **Guard:** frames clear of the burst vary by about 3 ms (26 B) at the 99th percentile, with a heavier tail of 1-2% at 25-35 ms for big frames. A guard of 10 ms covers most; 25 ms covers more; the rare 300 ms outlier was probably a host stall.

Boats that fit in a slotted schedule (one slot per boat per cycle)

Cycle	Rate	Bytes / slot	Packets	Slot, guard 10 ms	Boats	Slot, guard 25 ms	Boats
1 s	2 Hz	46	1	20 ms	46	35 ms	26

Cycle	Rate	Bytes / slot	Packets	Slot, guard 10 ms	Boats	Slot, guard 25 ms	Boats
1 s	5 Hz	110	2	30 ms	30	45 ms	20
1 s	10 Hz	214	3	43 ms	21	58 ms	15
2 s	2 Hz	84	1	21 ms	89	36 ms	52
2 s	5 Hz	214	3	43 ms	45	58 ms	33
2 s	10 Hz	420	5	67 ms	28	82 ms	23
4 s*	2 Hz	168	2	33 ms	117	48 ms	80
4 s*	5 Hz	420	5	67 ms	57	82 ms	46
4 s*	10 Hz	840	10	124 ms	31	139 ms	27

Model: 80 ms blocked per correction burst; a 2 s cycle means corrections every 2 s (one burst per cycle); *a 4 s telemetry cycle with corrections every 2 s has two bursts per cycle (160 ms blocked). Time on the air is estimated as 8 ms per radio packet plus 0.04 ms per byte (200 kb/s); the 8 ms is an upper bound from the measured fixed delay and has not been measured on the air. Boats are the number of slots in the cycle after the blocked time. The cost of a longer cycle is that a boat's newest position can be that old on arrival.

Caveats and next steps

- One Mac with two adapters. A Raspberry Pi adds its own serial/USB delay and variation; repeat the test on a Pi, and with the adapters that will be used.
- The conflict window is measured from the listener's decoded first correction message. A rover with a different adapter delay will see the same burst slightly earlier or later; the tracker should learn the offset from its own hearing.
- The rare long outliers (a 306 ms frame, a 273 ms burst) may be a host stall; not yet explained.
- Next: a burst tracker and transmit gate in the boat app using these numbers (anchor on the first correction message; block -40 to +20 ms plus a guard), tested on a Pi with the radio-latency script.

Appendix: raw reports (verbatim)

Run A - 26-byte frames, 300

Frames sent: 300 received: 293 lost: 7 (2.3%) frame size: 26 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 293 min 12.3 p50 13.0 p90 13.3 p99 36.1 max 45.4
jitter (p99 - p50): 23.1 ms spread (max - min): 33.1 ms

RTCM bursts heard: 84, every 1000 ms, up to 33 ms wide, up to 6 messages

near burst n= 16 min 12.3 p50 12.9 p90 34.8 p99 36.1 max 36.1 lost 7 of 23
clear n= 276 min 12.3 p50 13.0 p90 13.2 p99 15.9 max 45.4 lost 0 of 276

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 18, of which incomplete: 4
bursts with none: 66, of which incomplete: 3

Suggested slot guard: about 11 ms (= jitter 3.0 + anchor disagreement 3 + margin 5;

based on the "clear" frames). Treat it as a starting point: one host, two adapters, 300 frames.

Typical delay before a frame is heard: p50 13.0 ms

Run B - 84-byte batch frames, 600

Frames sent: 600 received: 589 lost: 11 (1.8%) frame size: 84 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 589 min 24.6 p50 25.1 p90 25.3 p99 49.1 max 306.5
jitter (p99 - p50): 24.0 ms spread (max - min): 281.9 ms

RTCM bursts heard: 164, every 1000 ms, up to 273 ms wide, up to 6 messages

near burst n= 178 min 24.6 p50 25.1 p90 25.4 p99 47.4 max 50.4 lost 7 of 185
clear n= 408 min 24.6 p50 25.1 p90 25.3 p99 49.1 max 306.5 lost 4 of 412

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 33, of which incomplete: 9
bursts with none: 131, of which incomplete: 6

Suggested slot guard: about 32 ms (= jitter 23.9 + anchor disagreement 3 + margin 5;

based on the "clear" frames). Treat it as a starting point: one host, two adapters, 600 frames.

Typical delay before a frame is heard: p50 25.1 ms

Run C - focus sweep, 26-byte frames, 150

Frames sent: 150 received: 149 lost: 1 (0.7%) frame size: 26 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 149 min 12.3 p50 12.9 p90 31.4 p99 48.2 max 52.5
jitter (p99 - p50): 35.3 ms spread (max - min): 40.2 ms

RTCM bursts heard: 155, every 1000 ms, up to 6 messages; width median 28 ms, 95th percentile 31 ms, widest 33 ms

near burst n= 48 min 12.3 p50 18.8 p90 38.2 p99 41.7 max 41.7 lost 1 of 49
clear n= 101 min 12.4 p50 12.9 p90 13.3 p99 48.2 max 52.5 lost 0 of 101

By when the frame was SENT, relative to the nearest burst start (10 ms buckets):

from..to ms	n	lost	p50 ms
-90..-80	3	0	12.7
-80..-70	5	0	12.9
-70..-60	6	0	13.1
-60..-50	7	0	13.0
-50..-40	8	0	12.9
-40..-30	2	0	48.2
-30..-20	7	0	36.8
-20..-10	8	0	32.1
-10..0	6	0	30.9
0..10	4	1	21.0
10..20	8	0	15.1
20..30	6	0	12.4
30..40	4	0	12.6
40..50	7	0	12.6
50..60	5	0	12.8
60..70	7	0	12.9
70..80	6	0	12.7
80..90	6	0	12.8
90..100	5	0	12.8
100..110	6	0	12.8
110..120	7	0	12.8
120..130	4	0	12.8
130..140	6	0	12.9
140..150	7	0	13.0
150..160	5	0	12.9
160..170	5	0	13.0

Sends from -40 ms to 10 ms around a burst start were lost or delayed - hold telemetry out of that range, plus a guard.

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 41, of which incomplete: 19
bursts with none: 114, of which incomplete: 2

Suggested slot guard: about 44 ms (= jitter 35.3 + anchor disagreement 3 + margin 5;
based on the "clear" frames).